FRACTION STUDIO PROPORTIONAL TOWERS

4-Way Multi-Notational Fraction Blocks (1/1 to 1/12)

CARD 04 / 10 • FRAC-01

MATHEMATICAL REFERENCE

STUDIO CONTINUUM: 1·Ten-Frames · 2·Dominoes · 3·Clock · **4·Fractions** · 5·Primes · 6·Algebra · 7·Probability · 8·Volvelle · 9·Quest · 10·Rack · 11·SC-32

1. WHAT IS THIS MANIPULATIVE?

PROPORTIONAL HEIGHT & 4-WAY MULTI-NOTATION

Fractions are proportional lengths on a 7mm grid.
Each block face shows fraction, decimal, and percent:
[1/3] (Symbol) = 0.333 (Decimal) = 33.3% (Percent)
Equal heights prove mathematical equivalence!

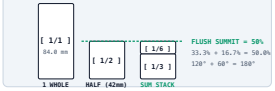
UNIVERSAL 84.00 MM WHOLE DATUM

- 1 Whole Datum: [1/1] = 12U = 84.00 mm reference rod
- Unit Base: [1/12] = 1U = 7.00 mm (Matches [2])
- Sub-Units: [1/2]=42mm, [1/3]=28mm, [1/4]=21mm
- Connectors: 1.85mm plug & recessed stacking socket

TWIN COMPARATOR WELL LID & STAGE

Drop fractional blocks into lid comparison stages; common denominators are visible instantly!

PROPORTIONAL EQUIVALENCE: 1/3 + 1/6 = 1/2



2. KIT PARTS INVENTORY (80 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
Whole [1/1]	1x	84.00 mm (12U = 1 Whole)	Classic White
Halves [1/2]	2x	42.00 mm (6U = 50.0%)	Warm Gold
Thirds [1/3]	3x	28.00 mm (4U = 33.3%)	Emerald Green
Fourths [1/4]	4x	21.00 mm (3U = 25.0%)	Royal Purple
Fifths [1/5]	5x	16.80 mm (2.4U = 20.0%)	Sky Blue
Sixths [1/6]	6x	14.00 mm (2U = 16.7%)	Coral Orange
Eighths [1/8]	8x	10.50 mm (1.5U = 12.5%)	Warm Gold
Tenths [1/10]	10x	8.40 mm (1.2U = 10.0%)	Sky Blue
Twelfths [1/12]	12x	7.00 mm (1U = 8.3%)	Slate Gray
Mini-Fractions	27x	7.00 mm Fractional Units	Assorted Family
3x1 Box & Lid	2x	49.0 mm Storage & Stage	Matte Charcoal

TOTAL: 78 Proportional Math Blocks + Storage Bin + Stage Lid = 80 Parts

FAMILY COLOR CODING SCHEME

- Color-Coded Denominators: Red=1, Orange=1/2, Green=1/3.
- Distinct Hues: Blue=1/4, Purple=1/6, Gold=1/12 unit base.
- Embossed Numerals: High-contrast white debossed numerals.
- Visual Verification: Identical colors share denominators.

3. WHY KEEP THIS REFERENCE CARD?

✓ NO LOST PIECE LEFT BEHIND

Classroom manipulatives get lost. Never abandon an incomplete set! Every individual math component is permanently stocked for single replacements.

Scan back QR or visit stackcalc.io/parts
Order individual replacement pieces anytime

✓ 10-SECOND SELF-CHECKING PACK-UP

Pack-up is a built-in math game! Stand each fraction family in its designated graduated slot. All full families level off at the 84.00 mm rim. A missing piece leaves an immediately visible gap. Comparison stage lid snaps flush for silent zero-rattle carry.

✓ HANDS-ON TUTORIAL ON REVERSE

Flip this card over for a complete step-by-step hands-on lesson with concrete numbers, stage arrangements, and calculator cross-references!

Keep this card stored inside the box lid!

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California, USA • Non-Toxic BPA-Free.
Classroom Lab Crates (10x / 30x Stations): stackcalc.io/schools

FUNDAMENTAL THEOREM OF RATIONAL ARITHMETIC

Proportional height is strictly preserved under common denominator fraction addition.

Universal 7.00mm Datum • Classroom Continuum (10 Kits)

OFFICIAL REFILL PORTAL

stackcalc.io/parts
Order replacement parts anytime

SERIES 1 • CARD 04 OF 10 • DOMAIN: PROPORTIONAL TOWERS & MULTI-NOTATION EQUIVALENCE • 84.0mm 1/1 WHOLE • 4-WAY COMPARISON FACES

LESSON GUIDE: COMMON DENOMINATOR ADDITION & EQUIVALENCE

Step-by-Step Hands-On Tutorial: 1/3 + 1/4 = 7/12 & 4-Way Multi-Notation

Grades 3–8 Fractions & Proportions • Hands-On Classroom & Home STEM Studio Series

CARD 04 / 10 • FRAC-01

MATHEMATICAL GUIDE

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN
stackcalc.io/products/fraction-studio/
Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.CONTENT.5.NF.A.1 (Unlike Denominators)
- CCSS.MATH.CONTENT.4.NF.A.2 (Fraction Comparisons)
- TEKS 111.7.b.3.H (Rational Operations)

Grades 3–8 Fractions & Proportions

MATHEMATICAL INQUIRY

Which smaller block family levels off flush with both fraction towers in the comparison slot?

1 STEP 1: SEAT 1/1 WHOLE IN REFERENCE WELL

Place the 84.00 mm Whole [1/1] block in the left reference slot.
Its 4 faces show: 1/1, 1.00, 100%, and 360°.
This physical block establishes the 100% benchmark height.

2 STEP 2: STACK 1/3 AND 1/4 BLOCKS

Place one [1/3] block (28.00 mm) in the right comparison well.
Stack one [1/4] block (21.00 mm) directly on top of the [1/3].
Total combined height: 28.00 + 21.00 = 49.00 mm.
Notice: neither thirds nor fourths alone measure this sum!

3-4 STEPS 3 & 4: TEST TWELFTHS & SOLVE

3. TEST COMMON TWELFTHS: Stack 7 twelfths: 7 × 7.00 mm = exactly 49.00 mm! Flush summit!
4. READ 4-WAY FACES: 0.333 + 0.250 = 0.583; 120° + 90° = 210°.
Calculator Verification: Press Fraction Display Icon ([FDISP] & [g][SIMP]) and run 1 [ENTER] 3 [+] 1 [ENTER] 4 [÷] [+] [FDISP] → 7/12.

5 CORE PRINCIPLE: COMMON DENOMINATORS ARE COMMON SIZES

You cannot directly add thirds and fourths because their blocks are different physical heights!
Converting to twelfths (4/12 + 3/12 = 7/12) literally swaps disparate pieces for identical 7.00 mm unit blocks so their heights can be counted directly!

★ EXTENSION INVESTIGATION: ADD: 1/2 + 1/3

Seat [1/2] (42mm) and [1/3] (28mm) together. Test sixths: how many [1/6] blocks reach the exact 70.0mm summit? (Scan QR for 3D animated solution!)
→ Assemble with your physical blocks, then scan the QR code to verify with the 3D step-by-step vector model.

STACKCALC ARCHIVAL INSTRUMENT LABS

Interactive Mathematics Manipulative Ecosystem.
Designed in Mountain View, California • Archival BPA-Free POM/PETG.

CROSS-REFERENCED CALCULATOR VERIFICATION

Every keystroke sequence verified with RPNCore CalculatorEngine.
Modular Hardware Continuum (9 Kits) • Universal 7.00mm Pitch • stackcalc.io

PARTS & SCHOOL REFILL PORTAL

stackcalc.io/parts
Classroom Lab Crates: stackcalc.io/schools

STACKCALC STEM PEDAGOGY & PHYSICAL COMPUTING

CONCRETE • REPRESENTATIONAL • ABSTRACT

1. TACTILE CONCRETE

7.00mm 1U PITCH
Every physical block dimension is locked to an exact integer ratio, providing tactile feedback of quantity.

MOTOR KINESTHETIC
Sliding, stacking, and interlocking builds spatial memory and intuitive number sense before symbols appear.

CONSERVATION OF QUANTITY
Mass, area, and volume remain invariant regardless of how blocks are grouped or rotated.

SELF-CORRECTING
Precision mechanical detents seat flush only when arithmetic identities and geometric proofs are correct.

CCSS.MATH.PRACTICE.MP5
Use appropriate tools strategically.

2. HARDWARE RPN STACK

POSTFIX ARITHMETIC
[ARG 1] [ENTER] [ARG 2] [OP]. Zero operator ambiguity, zero parentheses, and zero hidden precedence rules.

X, Y, Z, T REGISTERS
Every intermediate result is visible on the display stack. What is stacked on the desk is preserved in memory.

DIGITAL TWIN OF TOWERS
Physically stacking blocks onto a stage lid directly mirrors pushing values into the digital stack registers.

STEP-BY-STEP REASONING
Students observe each operator execute in real time, making compound calculations completely transparent.

CCSS.MATH.PRACTICE.MP1
Make sense of problems & persevere.

3. LEARNING CONTINUUM

GRADES K–2: FOUNDATION
Ten-Frames, Number Rack & Dominoes for subitizing, counting, 5-and-10 anchors, and place value fluency.

GRADES 3–5: FRACTIONS
Fraction bars, prime factorization towers, time gears, and discrete combinatorics with probability dice.

GRADES 6–12: ADVANCED
Algebraic factoring rails, unit circle volvelles, complex numbers, and professional scientific RPN workflows.

DIFFERENTIATED TIERS
Tactile physical manipulates bridge English language learners, visual learners, and accelerated STEM students.

CCSS.MATH.PRACTICE.MP4
Model with mathematics.

CLASSROOM CARE & SANITIZATION

1. FOOD-SAFE RECYCLED POLYMERS:
Manufactured from high-purity virgin polymer (PLA / PETG). Non-toxic, odorless, BPA-free, and compliant with RoHS, EN71-3, and ASTM F963 safety standards.

2. CLASSROOM CLEANING & DISINFECTION:
Wash blocks in warm water with mild liquid dish soap (up to 55°C / 130°F) or wipe with standard classroom disinfectant wipes. Allow to air dry flat before storage.

3. MODULAR SNAPFINITY STORAGE DETENTS:
Stackable 3x1 and 3x3 stage lids feature multi-axis detents tested to 10,000+ opening cycles. Trays pack rattle-free for classroom mobile cart transport.

4. SOCRATIC CLASSROOM DISCUSSION:
Encourage students to physically construct models to test hypotheses before executing calculator calculations: "What invariant property did your physical model prove?"

FRAC-01: IN-BOX SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC AUDIT CHECKLIST

[✓] MITUTOYO DIGITAL CALIPER DATUM:
All block dimensions conform to ±0.05 mm tolerance across universal 7.00 mm 1U visible height system.

[✓] 1.85 MM RETENTION SOCKET INTERFERENCE:
Stacking plugs seat flush with 0.15 mm diametral interference for zero-wobble tower verticality.

[✓] SNAPFINITY ZERO-RATTLE DETENT LOCK:
Multi-axis lid detents lock securely over 3x1 bin perimeter rim. Audible snap with 1.4 N retention.

[✓] CONTINUOUS 360° MICRO-FLUTING & SEAM TRAP:
2.0 mm pitch, 0.50 mm flutes. Slicer seam locked into rear-left corner seam trap notch.

[✓] FOOD-SAFE RECYCLED POLYMER:
100% virgin PLA / PETG. RoHS and EN71-3 compliant, non-toxic, odorless, child-safe.

[✓] IN-BOX COLLATERAL PLACEMENT:
Place this reference card inside the box lid or inside the storage tray.

RPN HARDWARE COMPATIBILITY

- Compatible with StackCalc SC-6 (Early STEM)
- Compatible with StackCalc SC-32 (Scientific RPN)
- Zero parentheses, zero operator precedence errors
- Physical towers are the exact digital twin of the stack!

TOTAL KIT INVENTORY: 80 PIECES

Verified by Assembly Tech & Machine Vision at Packing Bench.

Inspection Pass • Mountain View, CA

LIFETIME REPLACEMENT GUARANTEE

NO LOST PIECE LEFT BEHIND™

Classrooms lose blocks. We understand. Every individual manipulative block has a permanent part number. Visit stackcalc.io/parts to order exact replacement pieces with same-day dispatch ($1.99–$2.99/piece).

PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

- Net-30 Invoicing for US Public & Private School Districts.
- Volume Classroom Bundles (10-packs & 30-packs).
- Free Teacher Activity Books & Lesson Blueprints.
- Title I STEM & Carl D. Perkins Grant eligible.

Email: procurement@stackcalc.io
stackcalc.io/schools

3D INTERACTIVE VIDEO DEMOS

Scan the QR code on the back of this card (Side B) to access:

- 60fps 3D animated walkthrough of each lesson
- Interactive web 3D model with explode view
- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California